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Great Lakes ice classification using satellite C-band SAR multi-polarization data

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ABSTRACT

The objective of this study is to advance development of algorithms to classify and map ice cover on the Laurentian Great Lakes using satellite C-band synthetic aperture radar (SAR) multi-polarization data. During the 1997 winter season, shipborne polarimetric backscatter measurements of Great Lakes ice types, using the Jet Propulsion Laboratory C-band scatterometer, were acquired together with surface-based ice physical characterization measurements and environmental parameters, concurrently with European Remote Sensing Satellite 2 (ERS-2) and RADARSAT-1 SAR data. This fully polarimetric dataset, composed of over 20 variations of different ice types measured at incidence angles from 0° to 60° for all polarizations, was processed and fully calibrated to obtain radar backscatter, establishing a library of signatures for different ice types. Computer analyses of calibrated ERS-2 and RADARSAT ScanSAR images of Great Lakes ice cover using the library in a supervised classification technique indicate that different ice types in the ice cover can be identified and mapped, but that wind speed and direction can cause misclassification of open water as ice based on single frequency, single polarization data. Using RADARSAT-2 quad-pol and ENVISAT ASAR dual-pol data obtained for Lake Superior during the 2009 and 2011 winter seasons, algorithms were developed for small incidence angle (<35°) and large incidence angle (>35°) SAR images and applied to map ice and open water. Ice types were subsequently classified using the library of backscatter signatures. Ice-type maps provide important input for environmental management, ice-breaking operations, ice forecasting and modeling, and climate change studies.

Introduction

In the world's largest freshwater surface, covering an area of 245,000 km^2 with a drainage basin extending 1110 km north–south and 1390 km east–west (Goetz, 1990), ice cover in the Great Lakes is the most obvious seasonal transformation and inherently large-scale. The ice cover of the Great Lakes is a strong climate change indicator, and its seasonal change has a profound impact on regional environment, ecology, economics, navigation, and public safety. Knowledge of the ice cover and its interannual variability is critical for the assessment and prediction of the impact of the Great Lakes environment on socio-economic conditions. Therefore, it is necessary to identify and classify ice across the Great Lakes for use in many science studies and practical applications.

In this paper, we advance ice/water identification and ice classification algorithms using satellite synthetic aperture radar. We first discuss the importance of ice information then present the background of Great Lakes ice remote sensing. We present new advances in ice/water identification and classification algorithms using satellite SAR data with single and with multiple polarizations and show the results over several areas in the Great Lakes and then reach conclusions on utility and future studies.

Importance of ice information

Ice mass and water level change

Both ice cover and ice thickness are necessary to determine ice mass balance in lakes. Changes in lake ice mass directly affect hydrologic mass balance and lake water level. Furthermore, ice volume on the Great Lakes governs heat fluxes that impact regional water-mass balance such as lake-effect snow. Such an effect can result in anomalous snow accumulation on land. Record snowfall during the 1978–1979 snow season was 32.5 ft in Keweenaw County, Michigan. In the Lake Erie snowbelt, severe winter weather has a major impact on traffic flow maintenance on interstate highways and on public safety in general (Schmidlin, 1993). Lake ice mechanical strength also strongly depends on ice thickness, a very important navigation hazard for the shipping industry.

Water level in the Great Lakes can be significantly affected by the formation of ice jams. A naturally formed ice arch, which prevents ice transport into a river as a natural ice control structure, was investigated by Calkins et al. (1982). Large masses of thick ice can break the ice arch, causing an ice jam, and creating winter flooding. A particularly

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susceptible area is the St. Clair River where ice imported from southern Lake Huron creates large ice jams on the river. A record ice jam occurred in 1984 over a period of 24 days. A loss of $1.7 million/day (more than $3 million/day in today's value) in shipping loss was reported and it would take at least 3 years for the water levels in Lakes St. Clair and Erie to return to pre-jam conditions (Derecki and Quinn, 1986). Ice jams change river discharge and water flow which can have an impact on hydropower generation. Ice jam hazard conditions can be predicted if ice concentration is known. One study has found that the ice transport can be forecast based on the mean surface ice concentration, which correlates to the presence or absence of the ice arch corresponding to 9.5% and 27.3% ice concentrations, respectively (Daly, 1992).

Climate change

Global climate change can have a significant impact on the Great Lakes. Information on the ice cover and transport provides quantitative, sensitive, and early indicators to understand regional and global climate change. For example, lake ice-out date is considered a thematic climate indicator (Environmental Protection Agency, 2012). A number of studies have been commissioned to study climate change effects on the lakes (Smith, 1991). Ice cover duration is predicted by several General Circulation Models (GCM) to reduce by 5 to 13 weeks under the CO_2 (carbon dioxide) doubling scenario (Assel, 1991). Such a change is dramatic with respect to the already relatively short time of the seasonal ice cover duration currently in the Great Lakes.

Observations of ice seasons on the Great Lakes show an earlier ice-out indicating a trend toward earlier and warmer springs in the upper Midwest (Hanson et al., 1992). Freeze-up and break-up dates represent integrated climatic conditions during the winter–spring period when most warming is forecast to occur; this timing can provide an early indication of the climatic warming (Assel and Robertson, 1995). A study of atmospheric teleconnection patterns and winter severity over the Great Lakes basin suggests that two different subtypes of atmospheric circulation over the Northern Hemisphere determine the path of the polar jet stream and thus the winter severity over the Great Lakes (Rodionov and Assel, 2000). A recent study by Wang et al. (2012) found that Great Lakes lake ice mainly responds to the combined Arctic Oscillation and El Nino-Southern Oscillation patterns. Bai et al. (2012) also found that "both NAO and ENSO have impacts on Great Lakes ice cover." From data recorded between 1973 and 2010, Wang et al. (2012) found that there "was a significant downward trend in ice coverage from 1973 to the present for all of the lakes, with Lake Ontario having the largest, and Lakes Erie and St. Clair having the smallest."

Ecosystem change and effects

Ice cover and its movement impact regional ecosystems. Ice cover determines the transmittance of photosynthetically active radiation depending on snow cover, ice type, and ice thickness (Bolsenga et al., 1991). The duration of ice cover determines the amount of available light for overwintering algae, and is important to seasonal variations in algal mass (Adrian et al., 1995). A bloom of a typical winter–spring phytoplankton community was observed during ice cover (Vanderploeg et al., 1992). Winter ecology can be strongly affected by the ice cover reduction, which influences the environmental and habitat stability of the limnetic community (Crowder and Painter, 1991; Vanderploeg et al., 1992). In their investigation of offshore winter limnology in Lake Erie, Twiss et al. (2012) found that "winter blooms probably have consequences for summer hypoxia and how the lake responds to climate change." The change modifies and degrades the Great Lakes wetland and deep-water habitats of many species. Large concentrations of mallards, American black ducks, and mergansers are found on ice-free areas during winter (Prince et al., 1992). Ice freeze-up and ice break-up are related to the timing of seasonal drift and habitat change of *Lethocerus americanus* (Dubois and Rackouski, 1992). The number of days that ice cover exceeded 40% is an important input to the whitefish recruitment forecast model for northern Green Bay and North Shore areas of Lake Michigan (Brown et al., 1993). Knowing the ice cover percentage may therefore provide a quantitative forecast of fish recruitment for applications to marine resource management both by government authorities and by the fishing industry.

Background on Great Lakes ice mapping

Early work

Early investigations by various researchers were conducted to classify and categorize ice types and features (Chase, 1972; Bryan, 1975), to map ice distribution (Leshkevich, 1976; McMillan and Forsyth, 1976), and to monitor and attempt to forecast ice movement with remotely sensed data (McGinnis and Schneider, 1978; Rumer et al., 1979; Strong, 1973; Schneider et al., 1981). Most of the early classification and mapping research on Great Lakes ice cover was done by visual interpretation of satellite and other remotely sensed data (Rondy, 1971; Schertler et al., 1975; Wartha, 1977). Owing to the size and extent of the Great Lakes and the variety of ice types found there during the winter, the timely and objective qualities in computer processing of satellite data make it well suited for such studies.

Limitations of early sensors

Much of the satellite ice interpretation algorithm development in the Great Lakes region began during the Extension to the Navigation Season Demonstration Study conducted during the 1970s. However, many of the early studies were done by subjective interpretation of satellite and other remotely sensed data. Starting in the mid-1970s, a series of studies including field studies and computer digital image processing, explored techniques and algorithms to classify and map freshwater ice cover using LANDSAT and Advanced Very High Resolution Radiometer (AVHRR) optical data (Leshkevich, 1985) and later, ERS-1/2 and RADARSAT SAR data (described in Leshkevich and Nghiem, 2007).

During winter months, cloud cover and reduced hours of daylight over the Great Lakes impair the use of satellite imagery from passive sensors operating in the visible, near infrared, and thermal infrared regions. Passive microwave data acquired by satellite radiometers, such as the Special Sensor Microwave Imager (SSM/I) and the more recent Special Sensor Microwave Imager/Sounder (SSMIS), lack the spatial resolution required for monitoring and detailed analysis of Great Lakes ice cover. Although airborne high-resolution SAR or Side Looking Airborne Radar (SLAR) was used by the U.S. Coast Guard, the Canadian Ice Service and others for operational ice mapping, they were limited in their spatial and temporal coverage, were costly to operate, and were dependent on weather conditions for flight safety. The all-weather, day-and-night sensing capabilities, and large coverage of satellite SAR make it well suited to monitoring ice cover in the Great Lakes region.

The 1997 Great Lakes winter experiment (GLAWEX97)

For remote sensing of Great Lakes ice cover, a field experiment campaign (GLAWEX97) was conducted during the 1997 winter season across the Straits of Mackinac and Lake Superior. The campaign was coordinated in two expeditions on two different United States Coast Guard icebreaker vessels, the USCGC *Biscayne Ba*y in February and the USCGC *Mackinaw* in March. Aboard these icebreakers, the Jet Propulsion Laboratory (JPL) C-band polarimetric scatterometer was used to measure backscatter signatures of various ice types and calm open water at incidence angles from 0° to 60°. The radar measurements include incidence angles and polarizations of spaceborne synthetic aperture radars (SAR) on ERS-1 and 2, RADARSAT-1 and 2, ENVISAT, and the soon to be launched Sentinel-1 satellites.

The polarimetric scatterometer data together with in situ measurements yield a backscatter signature library that can be used to interpret SAR data for ice classification and mapping. Results were presented for backscatter signatures of Great Lakes ice types from thin lake ice to thick brash ice with different snow-cover and surface conditions (Nghiem and Leshkevich, 2007). The library was successfully used in the computer classification of calibrated satellite SAR data (Leshkevich and Nghiem, 2007). Computer analysis of ERS-2 and RADARSAT ScanSAR images of Great Lakes ice cover using a supervised classification technique indicates that different ice types in the ice cover can be identified and mapped, but that single frequency, single-polarization data can lead to misclassification of open water as ice at different ranges of incidence angle and wind conditions (Leshkevich and Nghiem, 2007). The different ice types are summarized in Table 1 and the ice properties and nomenclature are described in the published literature (Nghiem and Leshkevich, 2007).

Advanced algorithm development

Capability of multi-polarization data

The advancement in ice mapping with multi-polarization SAR data is based on the library of measured backscatter signatures for Great Lakes ice. The unique polarimetric backscatter signature library of different ice types and calm open water is directly applicable to RADARSAT-2 and Envisat data (same frequency band, all polarizations, and incidence angles) to continue development using their dual-polarization, quad-pol, and fully polarimetric capabilities.

One of the main goals of this study is to determine if multi-polarization data can be used to identify ice and open water without the ambiguity caused by variability in wind speed and direction over water. This ambiguity is a problem in using single polarization, single frequency data (Leshkevich and Nghiem, 2007). Multi-polarization SAR data provide greater discrimination and less ambiguity than single polarization data to improve ice type discrimination and mapping that are robust over a wide range of wind speed and direction.

To illustrate the capability of SAR data with multiple polarizations, we use Envisat Advanced SAR (ASAR) data acquired on 11 May 2003 over the Great Bear Lake located on the Arctic Circle in the Northwest Territories of Canada (Fig. 1, top panel). This dataset consists of backscatter data obtained by transmission at the vertical polarization and receiving the radar return at both vertical polarization (co-polarization *VV*) and horizontal polarization (cross-polarization *VH*).

In the single-polarization *VV* image (Fig. 1, lower left panel), the spatial pattern of the ice cover is seen with different backscatter levels in different areas across the lake surface. Note that the top right and lower left corner areas of the image are land. The *VV* image reveals two main ice packs separated by a refrozen lead running across the lake between the lower left land area and the upper right land area.

Table 1
Description of different ice types measured in the Great Lakes during GLAWEX97.

Ice type	Thickness	Surface condition
Brash ice (14)	Up to >5 m	Rubble field, crushed/broken ice plates, rough surface
Pancake ice (27)	15–18 cm	Some snow areas, very rough edge at rim of pancake ice
Stratified ice (21)	18–25 cm	Hard metamorphosed snow with mottled surface pattern, some relief
Lake ice w/crusted snow (10)	30–36 cm	Up to 5 cm snow cover, metamorphosed crusted surface
Rough consolidated ice floes (12)	>5 cm	Crescents of slush curd, mottled surface pattern, moist surface
New lake (black) ice (05)	2.5 cm	Dark surface with some pattern of melted, refrozen snow with relatively smooth surface

In the dual-polarization *VV*/*VH* image (Fig. 1, lower right panel), more details of different types of ice are identifiable compared to the single polarization *VV* image. This dual polarization image is obtained by mixing the *VV* and *VH* channels with red and blue for *VV* and green for *VH*. The refrozen lead between the two main ice packs appears distinctively in green in the *VV*/*VH* image while the backscatter level in this icy lead can have similar *VV* backscatter values as in other areas of the lake.

These results for the Great Bear Lake from Envisat ASAR show that multi-polarization SAR data are capable of detecting more ice types than using single polarization data. However, no surface truth was available to characterize properties of different ice types in the Great Bear Lake image. Moreover, backscatter change as a function of incidence angle needs to be considered to improve ice classification algorithms.

Incidence angle dependence

In different ranges of incidence angles from extended low (~10°) to extended high (~60°) SAR modes, the polarization signatures of multi-polarization backscatter have different behaviors. Therefore, various combinations of backscatter data at different polarizations need to be selected in lake ice mapping algorithms for different SAR modes at different ranges of incidence angle. In these algorithms, the first step is to identify ice and open water. In the next step, our library of C-band polarimetric backscatter signatures of different freshwater ice types (Nghiem and Leshkevich, 2007) can be applied to the imagery to classify and color-code the ice types. Once fully developed, the overall algorithm can be used in an automated procedure to obtain ice type and ice mapping.

The polarization ratio between two different co-polarizations, *HH* and *VV*, at C-band for the various ice types listed were derived for different ice types in the Great Lakes (Nghiem and Leshkevich, 2007). When backscatter is contemporaneously measured at dual co-polarization, we can obtain the co-polarized backscatter ratio defined as

$$r = VV/HH. \tag{1}$$

In the decibel (dB) domain, this ratio can be positive or negative because

$$r(\mathrm{dB}) = 10 \times \log(r) > 0 \ \text{ when } \ r > 1 \tag{2a}$$

$$r(\mathrm{dB}) = 10 \times \log(r) < 0 \ \text{ when } \ r < 1 \tag{2b}$$

where log is the common base-10 logarithm. For example, consolidated ice floes have a large positive ratio at incidence angles larger than 35°, indicating the characteristic of surface scattering. For pancake ice, the behavior of the co-polarized ratio is complex owing to complex structures in the surface and the volume of this ice type. The ratio of stratified ice is all negative over the range of incidence angles, indicating that the attenuation is low for this ice type and C-band electromagnetic waves can reach lower medium interfaces to enhance *HH* due to internal reflections. The backscatter ratio of lake ice with crusted snow is also negative or close to zero; however, the change of the ratio is opposite to that of stratified ice as a function of incidence angle.

The various ice types, except for new lake ice (backscatter typically below satellite SAR noise-floor), can be identified from open water over a wide range of wind speed and direction by using the copolarized ratio obtained from dual *HH* and *VV* polarization data at large incidence angles. This is because the set of curves of $r = VV / HH$ for open water under various wind conditions are well above the r curves for the ice types from the backscatter library (see Fig. 7, in Nghiem and Leshkevich, 2007). Furthermore, the full dataset of the polarimetric library contains all other polarization combinations for use in the development of lake ice classification algorithms using C-band multi-polarization data.

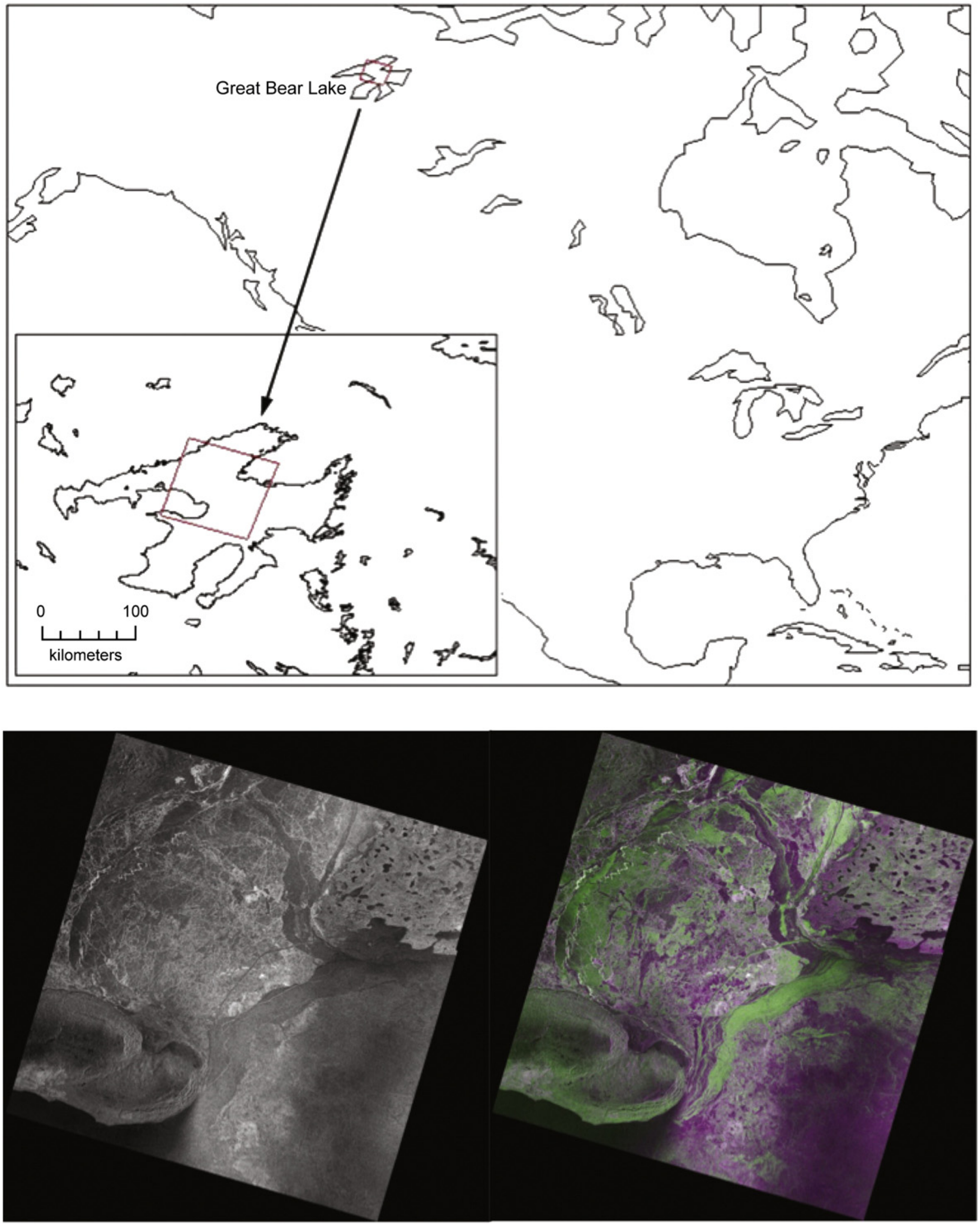

Fig. 1. ENVISAT ASAR images of ice cover in Great Bear Lake on 11 May 2003. Top panel shows the location of Great Bear Lake and the location of the ENVISAT ASAR scene. The lower left panel shows the *VV*-polarization image. The lower right panel is the image derived from a blending algorithm of *HH* and *HV* polarizations using ENVISAT dual-polarization data.

Advancing SAR algorithm development for different incidence angle ranges

Although the initial single-polarization algorithm validation showed that the algorithm correctly classified ice types in the ice-class library, open water was often misclassified owing to the ambiguity encountered in single polarization data due to variations in wind speed and wind direction over water. Therefore, it is necessary to advance the SAR algorithm for freshwater ice identification and classification using satellite C-band SAR data with multi-polarization, such as RADARSAT-2 polarimetric or quad-pol data and ENVISAT ASAR Alternate Polarization (AP) dual co-polarization data, to remediate this ambiguity.

Because of the incidence-angle behavior discussed above, the algorithm is developed for different ranges of incidence angles, over which radar scattering mechanisms have different characteristics depending on boundary conditions for transverse electric and transverse magnetic fields, and the relative volume and surface scattering at different polarizations. From our measured backscatter library together with the physical scattering characteristics, the algorithm is partitioned into two regimes: small incidence angles $\leq 35°$, and large incidence angles $>35°$. For each of these ranges of incidence angle, the algorithm is developed differently for ice and water identification owing to different scattering mechanisms at different incidence angles.

Table 2
Summary of SAR sensor and calibration characteristics.

Date	SAR sensor	Mode	Incidence angle (°)	Calibrated resolution (m)	Noise floor (dB)
3/18/09 (a)	RS2 Quad Pol[a]	FQ2	19–21.7	10 × 10	−33.5
3/20/11 (b)	RS2 Quad Pol[a]	FQ28	46–47	20 × 20	−33.5
4/6/11 (c)	Envisat ASAR AP[a]	IS6	39.61–42.76	60 × 60	−22.0

[a] See Radarsat-2 Product Description (2009) and Envisat ASAR AP Product Update (2009) for more details.

Satellite SAR data and image processing methods

Two RADARSAT-2 Quad-Pol scenes with concurrent in-situ data collected from the U.S. Coast Guard icebreakers and one ENVISAT ASAR AP scene were used in this study (see Table 2, Fig. 2). The RADARSAT-2 (SLC) and ENVISAT ASAR AP (ASA_APP) scenes were calibrated to radar cross-section (sigma nought) in the decibel (dB) domain using ENVI SARScape® software. Subsequent image processing and analysis were performed using the ENVI® image processing software. In the calibration process, the high-resolution orbital data was applied to the RADARSAT-2 FQ28 (3/20/11) scene and resolution was specified at 20 × 20 m.

Although a recent note from the Canadian Government (SOAR Program) advises of "a relative radiometric bias that occasionally affects RADARSAT-2 quad-pol products (Fine-Quad and Standard-Quad) [such that] channels with horizontal polarization on receive (*HH* and *VH*) have a radiometric bias of +0.5 dB relative to channels with vertical polarization on receive (*VV* and *HV*). The bias affects a small proportion (3%) of products acquired since the start of 2011 and prior to April 9, 2012." Upon checking per test described in the note, the RADARSAT-2 scene acquired for this study was found to be unaffected by this radiometric bias. The RADARSAT-2 FQ2 (3/18/09) scene was calibrated with native 5-meter resolution and 2 × 2 block averaged to reduce image size and help to reduce speckle (noise). The ENVISAT IS6 (4/6/11) scene was calibrated using 60 × 60 m resolution for the same reasons.

Ice/water "masks" were developed using the *HH* polarized scene for the small incidence angle (<35°) image (RS2-3/18/09) and the *HH* and *VV* polarized scenes (*VV*/*HH* ratio) for the large incidence angle (>35°) images (RS2-3/20/11 and ENVISAT-4/6/11). For small incidence angles it was found that using −3 dB as a threshold could separate ice and water in the *HH* imagery. For large incidence angles, a *VV*/*HH* ratio of 3.1 was used as a threshold to separate ice and water. Once ice cover was determined in the images, the calibrated ice backscatter library was used to classify the ice types at each incidence angle across the SAR (*HH*) image. The incidence angle of each pixel was determined using output from the calibration process recorded in an incidence angle file. The backscatter of each pixel was then compared to each ice type in the library at the same incidence angle to determine the ice type, and color-coded.

The RADARSAT-2 Quad-Pol Fine radiometric error is specified to be <1 dB in the RADARSAT-2 Product Description (2009). Because the overall uncertainty in calibration was estimated to be approximately ±1 dB, including both instrument and calibration uncertainty, the minimum and maximum signatures for each ice type were decreased and increased (respectively) by 1 dB to account for this uncertainty. Test cases without accounting for this uncertainty produced results with greater number of unclassified pixels and/or misclassifications.

Although our library of backscatter signatures includes the most common ice types found on the Great Lakes, it may not be all-inclusive resulting in some unclassified pixels. Using geographic information system (GIS) software, it was found that approximately 98% of the RADARSAT-2 small incidence angle (3/18/09) image was classified, 88% of the RADARSAT-2 (3/20/11) image, and 71% of the ENVISAT ASAR AP (4/6/11) image were classified. Validation was accomplished by comparing the in situ measured data (field notes and photos) with the classified output.

Results and discussion

Ice identification at small incidence angles

For small incidence angles, single-polarization (either *HH* or *VV*) SAR data can reasonably distinguish between ice and water at different wind speeds and directions (see Fig. 6, Nghiem and Leshkevich, 2007). This is because surface scattering dominates the scattering mechanism at small incidence angles where open water has high backscatter due to high relative permittivity (dielectric constant) of liquid water compared to that of ice on the lake surface.

For the small incidence angle range, the algorithm consists to two steps: (1) ice versus water identification, and (2) ice classification over areas identified as ice from the previous step. Both steps are based on the backscatter library (Nghiem and Leshkevich, 2007). To test this small-incidence algorithm, we use RADARSAT-2 SAR *HH* data collected on March 18, 2009 in Lake Superior as indicated by Scene 2 (Fig. 3, left panel). The range of incidence angles for this SAR scene is approximately 19° to 21°. This scene was collected by RADARSAT-2 SAR toward the west end of Lake Superior, and is located

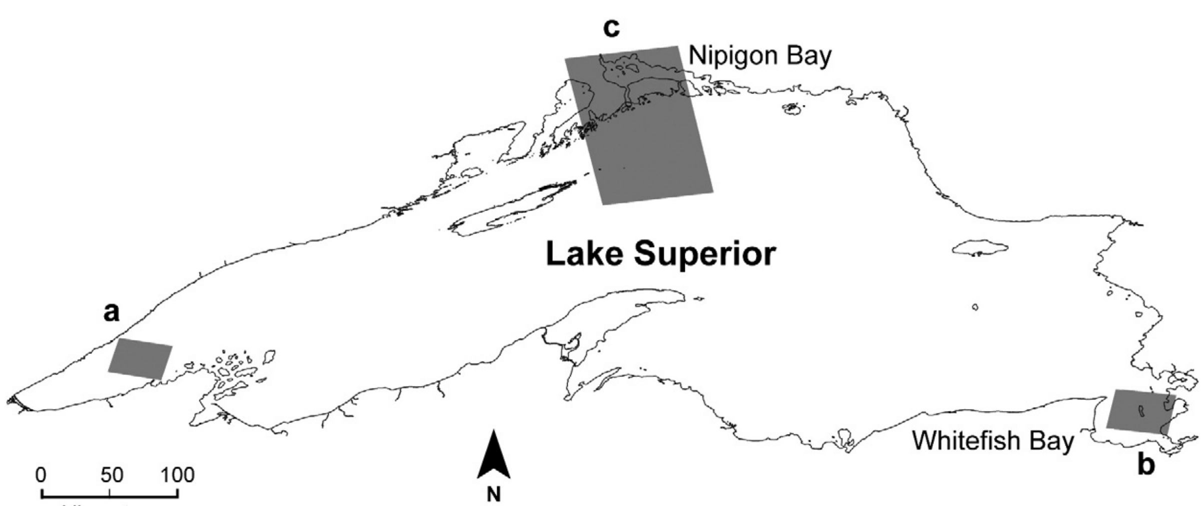

Fig. 2. Location of RADARSAT-2 Quad Pol and ENVISAT ASAR AP scenes on Lake Superior. (a) RADARSAT-2 small incidence angle scene, 3/18/09; (b) RADARSAT-2 large incidence angle scene, 3/20/11; (c) ENVISAT ASAR AP large incidence angle scene, 4/6/11.

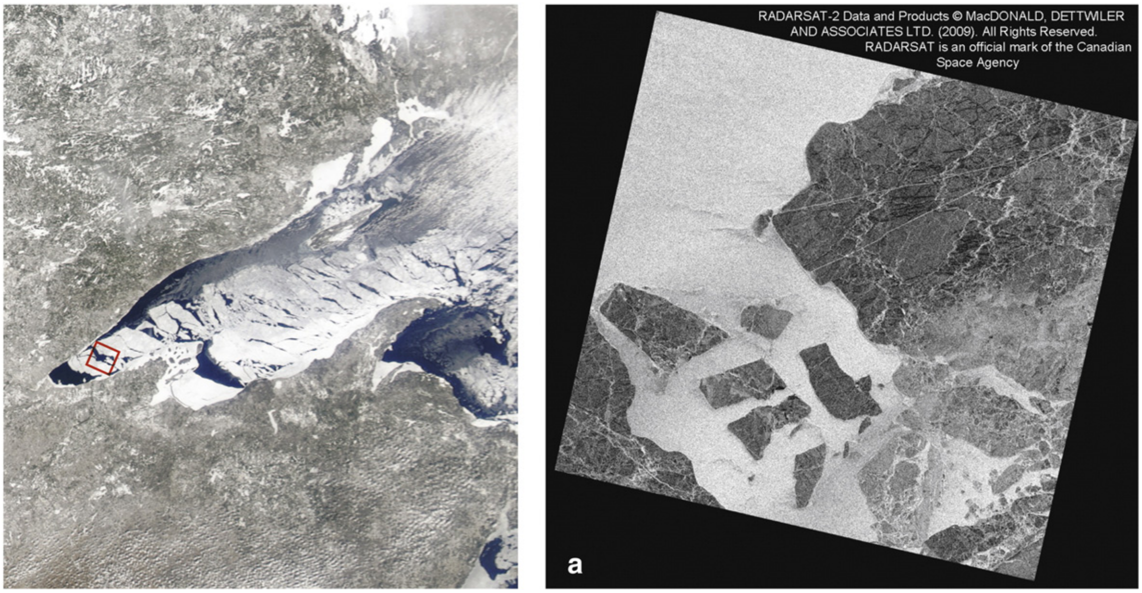

Fig. 3. RADARSAT-2 Quad Pol (*HH*) scene at small incidence angles collected on March 18, 2009 in Lake Superior. The left panel shows the location of the SAR scene in a MODIS true color image collected on March 13, 2009.

entirely over the lake surface. In this area, the ice pack was broken and several ice floes were detached and surrounded by open water (Fig. 3, right panel).

The ice algorithm results are shown in Fig. 4 for both ice/water identification (left panel) and ice classification (right panel). Results from the identification algorithm show distinctive areas of ice and water. In the large ice pack in top right portion of the left panel in Fig. 4, the two long linear features are two ship tracks. These tracks were created by Coast Guard ice-breaking operations, usually carried out in mid to late March. Many small leads which are fractures in the ice with open water are found not only in large ice packs but also in several ice floes.

While the large ice floes are clearly visible, the ice identification algorithm can also detect many small ice parcels of different sizes (lower right portion in the left panel, Fig. 4). However, in the open water areas, a number of pixels are misidentified as ice because of noise in SAR measurements at a high spatial resolution. These are approximately 11% of the number of the overall water pixels, and can be disregarded. The backscatter library suggests that the success rate of ice identification is higher for smaller incidence angles (<30°), which explains the good result in this case because of the range of small incidence angles between 19° and 21°.

After identifying ice and open water, we use the resultant ice mask and apply the classification algorithm (Leshkevich and Nghiem, 2007) across the ice areas. Results of ice classification (right panel, Fig. 4) reveal more brash ice and pancake ice (due to wind and wave actions) in the lower right portion of the SAR image. The ice in this area was in a complex matrix of different floe size indicating that this ice had experienced severe ice deformation consistent with the formation of brash ice. Nevertheless, there are isolated pixels in the open water that are misclassified as brash ice (red pixels) due to noise in the SAR measurements. "Moving window averaging" may reduce the noise and improve the ice classification; however, this method would reduce the spatial resolution and increase the mixed pixel problem at the ice edge.

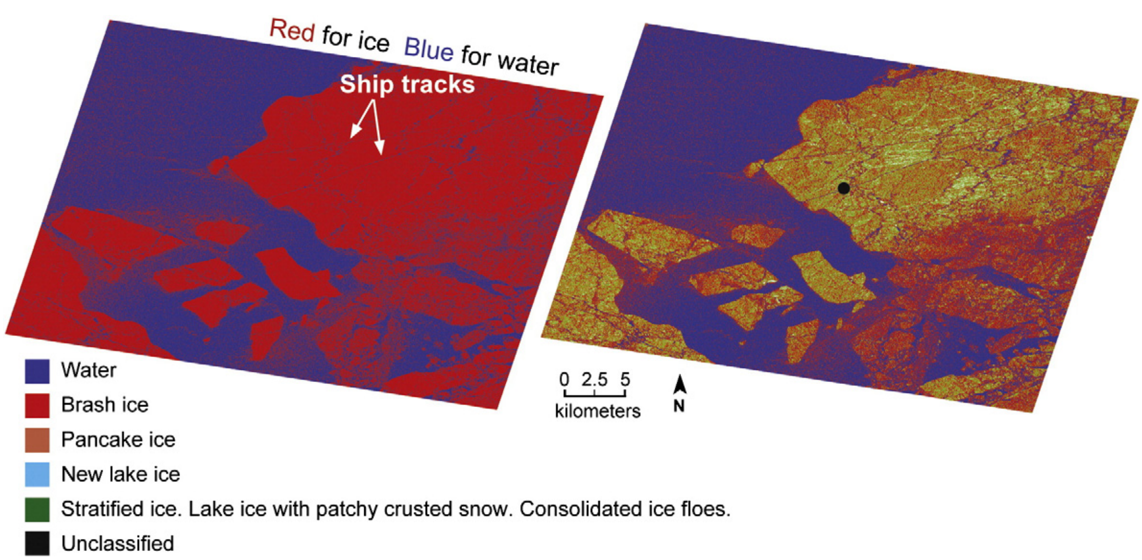

Fig. 4. Results for ice versus water identification (left panel) and ice classification (right panel) for the RADARSAT-2 Quad Pol (*HH*) scene at small incidence angles collected on March 18, 2009 in Lake Superior. Note that the algorithm is capable of detecting the ship tracks as marked by the white arrows in the ice-identification map. The black dot marks the approximate location from which the photo in Fig. 5 was taken.

From field observations along the ship tracks (Nghiem and Leshkevich, 2011), a majority of the ice types in this section of Lake Superior consisted of lake ice with crusted snow, consolidated ice floes, and stratified ice, which are among the most common ice types in the Great Lakes. Those types are shaded in green in the right panel of Fig. 4. For example, on March 17, 2009, along the ship track near the middle of the scene (47° 02.357′ N, 91° 20.830′ W), the ice consisted of lake ice (also known as black ice) with small (some large) broken pieces of ice frozen in the lake ice with windrows of crusted snow and some ridging (Fig. 5). Air temperature was 48 °F. Although in situ observations were not directly available for the ice floes in the lower left quadrant of the image (right panel, Fig. 4), these floes broke off from the main ice pack (the one with the ship tracks), thus we expect the ice types to be similar to those on the main ice pack. This is consistent with the green shade on those ice floes in the ice classification image. The limitation of SAR (*HH*) data at small incidence angles is that the backscatter ranges of these ice types overlap, and thus backscatter data at small incidence angles cannot fully resolve and separate each of these ice types individually if grouped together.

When zooming into the image, a number of pixels appear in black, which represents unclassified ice types. This is because the backscatter library does not capture all ice types found across the Great Lakes, although it includes the main types of ice usually encountered in the Great Lakes. Furthermore, there can be a mixture of different ice types in a given pixel (mixed pixel) in the SAR image where the resultant backscatter may not fit an ice type within the backscatter library used in this ice classification algorithm. Overall, this example shows that C-band SAR data at single polarization are useful to identify and classify ice over the range of small incidence angles, although the number of separable ice types is limited.

Ice identification at large incidence angles

For large incidence angles, dual-polarization (*VV* and *HH*) SAR data are necessary to distinguish between ice and water at different wind speeds and directions (see Fig. 7 in Nghiem and Leshkevich, 2007). This is because backscatter from open water is dominated by rough surface scattering with larger values of $r = VV / HH$, and backscatter of ice is mainly from volume scattering with smaller value of the co-polarization ratio r (Nghiem and Leshkevich, 2007).

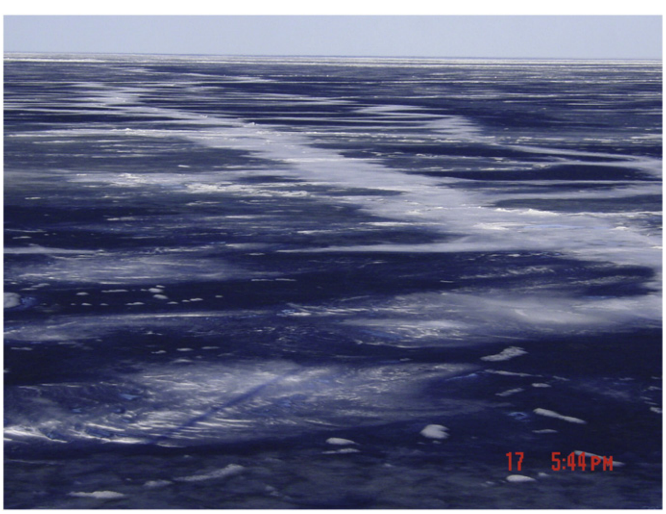

Fig. 5. Photo of the ice cover taken from the USCGC Alder on March 17, 2009 looking north-east showing the patchy crusted snow remaining from the melting of the snow windrows. Some of the melted snow (slush) can be seen refrozen into the surface.

In the range of large incidence angles, the algorithm also consists of two steps: (1) ice versus water identification using the *VV*/*HH* ratio library, and (2) ice classification over areas identified as ice from the previous step. In step one, except for calm water or new ice for which the backscatter signature is near or below the noise floor, ice and water are expected to be well separable. In step 2, the ice classification is based on *HH* polarization (*VV* can also be used) using the backscatter library (Leshkevich and Nghiem, 2007).

RADARSAT-2

To test the large-incidence algorithm, we use RADARSAT-2 SAR quad-polarization data collected on March 20, 2011 in Whitefish Bay, Lake Superior (Fig. 6), where the SAR scene is presented only with the *HH* polarization for incidence angles from 46.0° to 47.3°. Ile Parisienne is seen in the middle of the image, and some land areas are located along the right side of the SAR scene. The main ice pack resided mostly in Canadian waters on Whitefish Bay (right-hand side in Fig. 6). On the U.S. side, there was a collection of broken ice floes with different sizes.

The ice/water identification results of both the main ice pack on the Canadian side and the ice floes on the U.S. side are identified well in the SAR image (Fig. 7). Although land masked, Ile Parisienne appears as "ice" from the identification algorithm because it has similar co-polarization signature as ice and is identified as non-water. However, there are a number of pixels that are open water and misclassified as ice and vice versa. These pixels tend to be isolated ice pixels in an open water area (or water pixel in an ice area). These misclassified pixels are caused by noise in both the *VV* and *HH* channels of the SAR data, which translate into spurious errors in the co-polarization ratio r. Overall, the ice and water identification is correct and the use of dual co-polarization SAR data is suitable for the range of large incidence angles.

The ice classification results (Fig. 7) compare quite well with field observations obtained in GLAWEX 2011 (Nghiem and Leshkevich, 2011).

The ice identification and classification maps are geolocated and can be overlaid on Google Earth. Thus, these maps are ready for use together with the Global Positioning System (GPS) to locate ship tracks on the ice maps. For example, GPS data along the USCGC *Mackinaw* icebreaker track can be imported in Google Earth to plot out on the ice maps either in real time directly from a GPS device or from an existing GPS data file (Google Earth, 2011). Once the end-to-end processor is automated in a SAR tool-box to input, calibrate, geolocate, and process multi-polarization satellite SAR data into maps of ice types in the Great Lakes using the advanced multi-polarization SAR algorithm, the ice maps will be directly applicable to support USCG icebreaking operations and maritime navigation for the commercial shipping industry as well as scientific research and field experiment campaigns.

The land-fast ice between Ile Parisienne and the Canadian mainland is usually stable throughout the season. On March 24, 2011, the USCGC *Mackinaw* was heading north in the fast ice to the east of Ile Parisienne (ship track shown in Fig. 7 right panel). The photo in Fig. 8 was taken at approximately 10:15 am (46° 38.690′ N, 84° 38.719′ W) looking west at the south end of Ile Parisienne. The air temperature was −14.7 °C. The ice cover consisted of lake ice (also known as black ice) and new ice with remnants of windrowed snow with variable amounts of snow ice. There were areas of broken and refrozen pieces of ice. Ice thickness varied in the area with thicknesses up to about 30 cm with areas of snow ice varying from 0 to 7.6 cm. It is speculated that the warm temperatures on March 16 and 17 (7 °C and 12.8 °C, respectively contributed to melting of the windrowed snow creating areas of new ice or refreezing on the lake ice when the temperatures dropped to −5.0 °C by March 19 and reached −15 °C by March 24. The relatively calm water between the ice floes and the land-fast ice likely produced new ice or thin ice as shown color-coded cyan on the ice classified image in Fig. 7.

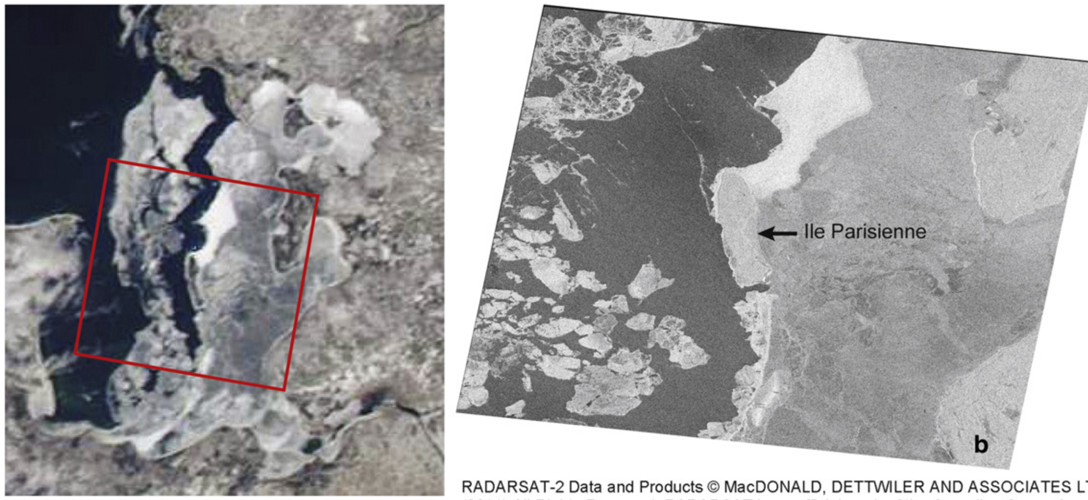

Fig. 6. The left panel shows a MODIS true color image of Whitefish Bay collected on March 19, 2011, and the red box outlines the RADARSAT-2 Quad Pol scene (right panel) shown with *HH* polarization only collected on March 20, 2011. Note the ice floe separating and moving westward from the land-fast ice east of Ile Parisienne.

ENVISAT ASAR AP

To further test the large incident angle ice/water algorithm with Envisat Alternating Polarization (AP) data, we obtained an Envisat ASAR scene of Nipigon Bay in northern Lake Superior taken on April 6, 2011. Although there is no in situ data for this scene, from past field work on adjacent Thunder Bay at about the same time of the season, we feel confident that the bay was predominantly covered with lake ice having a snow ice layer covered with snow, similar to the ice cover that we measured in 2009 on Thunder Bay (which had a thickness of up to 76.2 cm). Once ice forms on these northern Lake Superior bays, it has been found to remain fairly stable (fast) and consistent throughout the winter as seen in MODIS true color imagery throughout the season. After applying the ice/water mask to this large incident angle scene, the library of calibrated backscatter signatures was applied to the scene to classify the ice types (Fig. 9).

The relatively large percentage of unclassified (black) pixels in this scene would likely indicate an ice type that is not in the library of ice backscatter signatures. The backscatter values of these unclassified pixels can be derived. However, with no validation measurements for this scene, ice type identification would be difficult, although one could speculate a thicker snow ice layer based on the higher (brighter) backscatter.

Conclusions

We have successfully improved a SAR algorithm for ice classification and mapping over the Great Lakes using satellite multi-polarization SAR

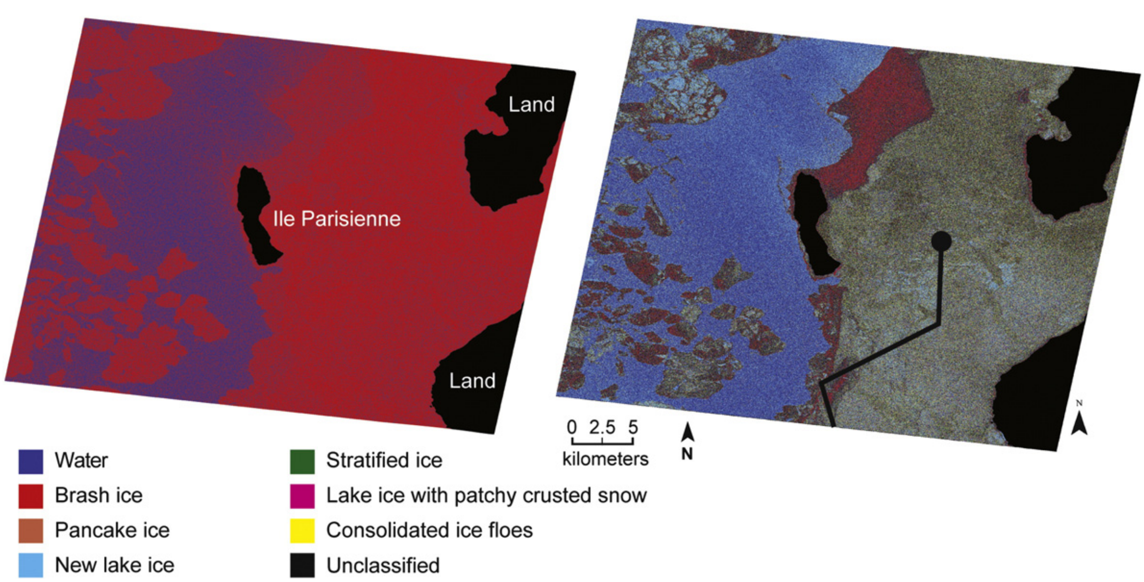

Fig. 7. Results for ice and open water identification (left panel) using the co-polarization ratio $r = VV / HH$ from the RADARSAT-2 SAR scene at large incidence angles collected on March 20, 2011 over Whitefish Bay in Lake Superior. Red is for ice and blue for open water. The large open water between the ice floe and the land-fast ice is clearly seen. The right panel shows the results of the ice signature library applied to the *HH* scene (ice classification) once water is identified. The black line represents the approximate ship track of the USCGC *Mackinaw* in the ice east of Ile Parisienne on March 24, 2011 where "ground truth" data was collected (black dot marks location of photo in Fig. 8).

Fig. 8. Photo of the ice cover taken from the USCGC *Mackinaw* on March 24, 2011 looking westward toward the southern end of Ile Parisienne. Notice the remains of wind rowed snow after recent melting.

data. Results are obtained from both RADARSAT-2 and Envisat AP ASAR data, which compare well with field observations. To avoid the misclassification of open water as ice that can result from the effects of wind speed and direction, we have developed two algorithms based on incident angle to identify ice and open water. For small incident angles (<35°) only the *HH* polarized scene is needed. For large incident angles (>35°) both *HH* and *VV* polarizations are needed to derive a *VV*/*HH* ratio from which the ice/water mask is created. Once ice and open water were identified, we applied the calibrated ice backscatter library to the *HH* polarized image to classify and map the ice types in the scene. For the small incident angle data, clarity improved when three of the classified ice types were mapped (color-coded) the same, owing to the overlapping backscatter signatures from these types (lake ice with crusted windrowed snow, rough consolidated ice flows, and stratified ice) in *HH* polarized data. However, at small incident angles, these types can be better differentiated in *VV* polarized data and this needs to be further investigated for possible algorithm improvement. For *HH* polarized data at large incident angles, these types can be well identified. For current operational ice mapping at the National Ice Center and Canadian Ice Service, C-band SAR (mostly *HH* polarized) data is visually interpreted and analyzed. Our freshwater ice classification algorithm can be automated to help with ice/water identification, determination of percent ice cover, and to classify and map ice types, which is important for operational ice breaking activities in support of winter navigation as well as for winter ecology, fisheries, and climate studies.

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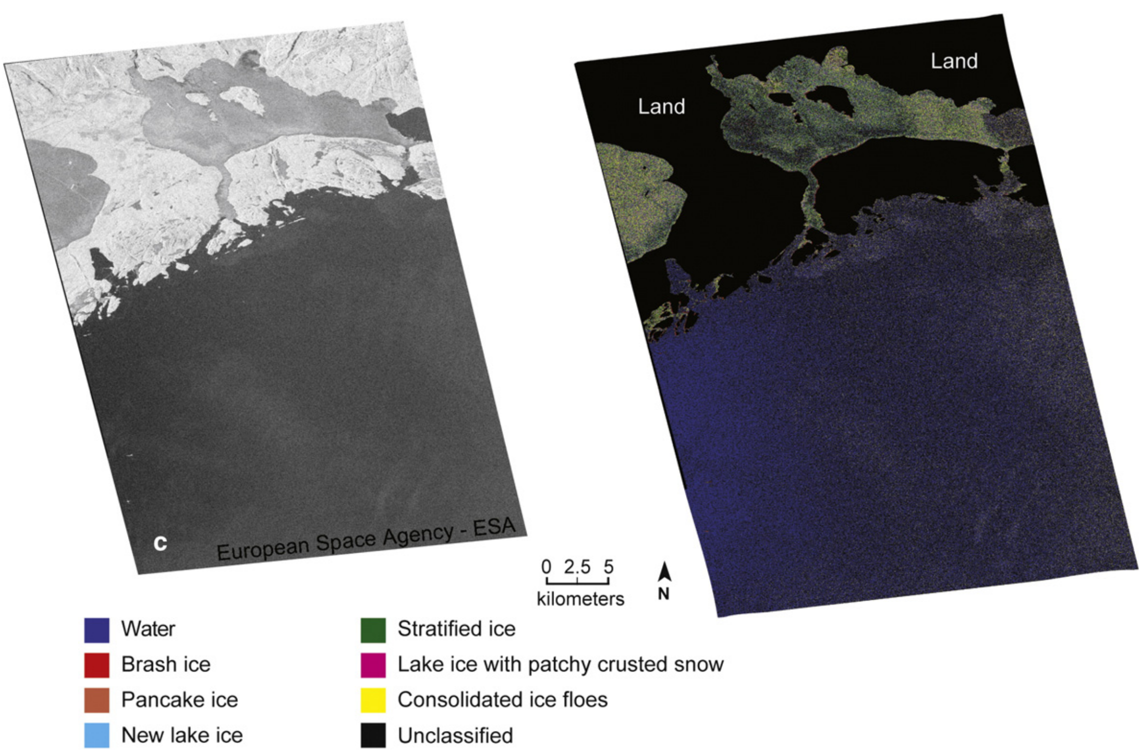

Fig. 9. ENVISAT ASAR AP large incidence angle scene (*HH* polarization) collected on April 6, 2011 over Nipigon Bay, Lake Superior (left panel). Results of ice classification for the ENVISAT ASAR (IS6) *HH* scene at large incidence angle (right panel).

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